

Distributed Control and Communication Framework for Connectivity Preservation in Flow Driven Robot Fleets

Abstract—We consider deployment of underwater robot fleets in flow dominated environments where communication is short range and unreliable, localization may be denied, and actuation energy is limited. The key systems challenge is to maintain end-to-end connectivity under flow-driven dispersion while executing coverage and target-discovery objectives under realistic communication constraints.

We propose a two layer architecture. First, a distributed topology maintenance layer sparsifies the time varying communication graph to a connected, spanning tree like subgraph *when the pruning epoch converges*, using only local neighbor exchanges. In contrast to communication heavy graph methods, the topology layer relies on constant size messages and local parent selection to prune redundant edges; we explicitly account for bytes per message, messages per update, and delivered bandwidth and evaluate robustness under packet drops, random delays, asynchronous broadcasts, and neighbor churn. Second, each robot solves a standard controlled Lyapunov function (CLF) and controlled barrier function (CBF) with a quadratic program (QP) that minimally modifies a nominal goal seeking controller to satisfy collision avoidance and range maintenance constraints on the maintained edges.

We validate the complete pipeline in 3D HoloOcean simulations with multiple BlueROV2 vehicles under an advection-diffusion motion model, demonstrating coverage-driven exploration and goal-directed navigation while preserving end-to-end connectivity with measured communication budgets under impaired channels.

I. INTRODUCTION

We target deployment of underwater robot fleets in large-scale environments (e.g., deep ocean, under-ice cavities) where localization is denied, communication is intermittent and bandwidth-limited, and transport is dominated by time-varying flows and eddies [1]–[3]. In our setting, robots are released to drift and disperse to increase sensing footprint (coverage), then reconfigure toward detected targets without losing connectivity.

Our approach releases robots together and exploits flow-driven drift, applying control only when needed to keep robots within prescribed bounds and to prevent loss of connectivity, enabling dispersion with connectivity and subsequent reconfiguration toward emerging targets [4]. This is supported by prior results showing flow-exploiting, intermittently actuated surface vehicles [5], [6] and by standard drift-diffusion models for transport in water [7]–[9] that match observed underwater robot motion [10], [11]. Although dispersion promotes coverage of large task spaces, two challenges remain open: maintaining end to end connectivity under flow driven neighbor churn, and enabling goal driven reconfiguration using only localized information and unreliable, bandwidth limited communication. Prior work on

multi robot connectivity has emphasized graph theoretic formulations [12] and control barrier function (CBF) approaches for energy efficient constraint enforcement. A representative direction maintains connectivity through spectral quantities such as the Fiedler value [13], which can be effective but typically requires global information or centralized computation and is expensive to maintain online. Distributed CBF formulations avoid global spectral computations by enforcing local pairwise range constraints [14], [15]; however, applying constraints to all instantaneous neighbors can be communication and control intensive in dense graphs, especially when the neighbor set changes under dispersion.

To reduce this burden, recent hybrid frameworks restrict connectivity maintenance to a sparse coordination structure, e.g., using a minimum spanning tree (MST) and then applying local CBF constraints only on selected links [16]. This idea is promising for scalability, but constructing such sparse structures is classically communication-intensive in fully distributed settings: exact distributed MST protocols require substantial coordination and message complexity [17], [18], and practical alternatives trade optimality for reduced overhead [19]. For our setting, the key systems constraint is not the optimality of the sparse topology but the feasibility under realistic underwater communication constraints and channel impairments. Recent progress has shown that connectivity relevant structure can be inferred using distributed procedures with limited information exchange [20]. Motivated by these observations, we construct a spanning tree coordination structure using constant size neighbor broadcasts and local parent selection, and we explicitly quantify bytes per message, messages per epoch, and delivered bandwidth under packet drops, random delays, asynchronous updates, and neighbor churn.

Upon the maintained spanning-tree structure, we apply a standard CLF-CBF-QP safety filter.

- A distributed δ -BFS pruning protocol that constructs a spanning-tree-like coordination topology *when an epoch converges* using constant-size broadcasts, with explicit payload accounting and measured tree success rates under packet drops, random delays, asynchronous updates, and neighbor churn.
- A two-layer architecture that couples topology maintenance with CLF-CBF control to support coverage-driven exploration and goal-reaching while enforcing collision avoidance and range maintenance only on maintained edges.
- Evaluation in the 3D HoloOcean [21] simulator with BlueROV2 [22] vehicles under an advection-diffusion

motion model, reporting pruning success, convergence time, and delivered bandwidth alongside closed-loop coverage/goal-reaching performance.

Our experiments show that δ -BFS achieves reliable spanning-tree formation under packet drops, bounded delays, and asynchronous broadcasts, while requiring substantially less communication and producing sparser maintained topologies than an adjacency-consensus baseline; we further demonstrate end-to-end goal reaching with connectivity preservation in 3D HoloOcean.

This paper is organized as follows. Sec. II introduces the problem and modeling assumptions. Sec. III presents the distributed topology maintenance protocol and the CLF CBF control layer. Sec. IV describes the communication impairments and experimental setup and reports results from communication stress tests and 3D HoloOcean simulations. Sec. V concludes and outlines limitations.

II. BACKGROUND AND PROBLEM FORMULATION

We now formalize the problem. This section introduces the robot dynamics model, states the assumptions, defines the graph theoretic notation, and presents the formal problem statement.

A. Advection-diffusion model [7]–[9]

We model a fleet of N robots operating in $\mathbb{R}^n$ (with $n = 2$ used for visualization in our simulations) released together in the water. Their collective motion can be decomposed into a drift component induced by an ambient flow field and a stochastic dispersion component due to unresolved turbulence and small perturbations. Let $x_i(t) \in \mathbb{R}^n$ denote the position of robot i at time t , where $i \in \{1, 2, \dots, N\}$, and let

$$\bar{x}(t) = \frac{1}{N} \sum_{i=1}^N x_i(t)$$

be the center of mass of the group.

The evolution of the center of mass is governed primarily by the advective flow field, i.e.,

$$\dot{\bar{x}}(t) \approx f_{\text{flow}}(\bar{x}(t), t),$$

where f_{flow} is the background velocity field. Each robot then deviates from $\bar{x}(t)$ due to unresolved small-scale turbulence and fluctuations, which we model via Brownian motion:

$$\begin{aligned} dx_i(t) &= (f_{\text{flow}}(x_i(t), t) + v_i(t)) dt + \sigma_{\text{diff}} dW_i(t), \\ \dot{v}_i(t) &= u_i(t) - \gamma v_i(t), \end{aligned} \quad (1)$$

where $v_i(t) \in \mathbb{R}^n$ is the robot velocity, $u_i(t) \in \mathbb{R}^n$ is the thrust input, $\gamma > 0$ is a velocity damping factor, and $W_i(t)$ is a Wiener process capturing diffusion with intensity σ_{diff} .

Thus the group drifts with the flow through the motion of $\bar{x}(t)$, while individual robots spread stochastically around the drifting center due to the independent diffusion terms.

Discrete-time simulation. In simulation we integrate (1) using Euler–Maruyama with step Δt :

$$\begin{aligned} x_i^{k+1} &= x_i^k + (f_{\text{flow}}(x_i^k, t_k) + v_i^k) \Delta t + \sigma_{\text{diff}} \sqrt{\Delta t} \xi_i^k, \\ \xi_i^k &\sim \mathcal{N}(0, I), \end{aligned} \quad (2)$$

together with the velocity update $v_i^{k+1} = v_i^k + (u_i^k - \gamma v_i^k) \Delta t$. This makes the diffusion term an explicit part of the state evolution used in our experimental evaluation.

We assume the ambient flow is spatially smooth and Lipschitz over the operating domain, there exists $L_f > 0$ such that

$$\|f_{\text{flow}}(x, t) - f_{\text{flow}}(y, t)\| \leq L_f \|x - y\|$$

for all relevant x, y, t . This captures non uniform flows where relative drift generally does not cancel.

B. Control-affine model

For controller design we use a reduced control-affine representation of the position dynamics,

$$\dot{x}_i(t) = f_{\text{flow}}(x_i(t), t) + u_i(t) + d_i(t), \quad \|d_i(t)\| \leq \bar{d}, \quad (3)$$

where $u_i(t)$ is the commanded control input and $d_i(t)$ aggregates unmodeled effects into a bounded disturbance. We emphasize that (3) is used as a *design abstraction* for the CLF–CBF safety filter, while our simulations evolve robots under the advection–diffusion model in (2).

Because the flow is evaluated at each robot’s position, it does not cancel in relative coordinates when the flow is spatially varying. Defining $z_{ij} \triangleq x_i - x_j$, the relative dynamics satisfy

$$\dot{z}_{ij} = \Delta f_{\text{flow}}(x_i, x_j, t) + (u_i - u_j) + (d_i - d_j), \quad (4)$$

with $\Delta f_{\text{flow}}(x_i, x_j, t) \triangleq f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)$.

Under intermittent communication, robot i does not have access to the instantaneous neighbor control $u_j(t)$. Following Assumption 3, we write

$$u_j(t) = \hat{u}_j(t) + e_{ij}(t), \quad \|e_{ij}(t)\| \leq \bar{e}, \quad (5)$$

where $\hat{u}_j(t)$ is the most recently received neighbor control estimate and $e_{ij}(t)$ capture staleness. Substituting (5) into (4) yields the implementable form used in the robust CBF constraints:

$$\dot{z}_{ij} = \Delta f_{\text{flow}}(x_i, x_j, t) + (u_i - \hat{u}_j) + (d_i - d_j) - e_{ij}. \quad (6)$$

So, the flow contribution cancels only in the special case of spatially uniform flow. In our setting, spatial variation of f_{flow} is explicitly bounded via the Lipschitz constant L_f , which enters the connectivity and invariance analysis.

The time derivative of the squared inter-robot distance can be written as

$$\frac{d}{dt} \|z_{ij}\|^2 = 2z_{ij}^\top \left(\Delta f_{\text{flow}}(x_i, x_j, t) + (u_i - \hat{u}_j) + (d_i - d_j) - e_{ij} \right),$$

which makes explicit that the flow contributes through the *difference term* $\Delta f_{\text{flow}}(x_i, x_j, t) = f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)$.

Notice that the choice of a two-dimensional environment is made solely for ease of visualization and controlled parameter sweeps. The developed method is dimension-agnostic; end-to-end validation is performed in the 3D HoloOcean simulator [21] using the BlueROV2 vehicle model [22].

C. Assumptions on communication, control and connectivity

Assumption 1. At time t , the communication network is an undirected disk graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ with range $R_{\max} > 0$:

$$(i, j) \in \mathcal{E}_t \iff \|x_i(t) - x_j(t)\| \leq R_{\max}. \quad (7)$$

Each robot i communicates only with one-hop neighbors

$$\mathcal{N}_i(t) \triangleq \{j \in \mathcal{V} : \|x_i(t) - x_j(t)\| \leq R_{\max}\}, \quad (8)$$

(with no central coordination) and has a unique identifier; $\mathcal{G}_0$ is connected. For all i, t : $\|u_i(t)\| \leq u_{\max}$, $\|d_i(t)\| \leq \bar{d}$. The flow field $f_{\text{flow}}(x, t)$ is spatially Lipschitz with constant L_f over the operating domain.

Assumption 2. There exist margins $\varepsilon_s > 0$ and $\varepsilon_c > 0$ such that

$$0 < d_{\min} + \varepsilon_s < R_{\max} - \varepsilon_c,$$

and the available control authority satisfies

$$u_{\max} > \bar{d} + L_f R_{\max}.$$

Assumption 3. For each maintained link $(i, j) \in \mathcal{E}_{\text{topo}}(k)$, robot i receives from j a timestamped packet containing its most recently applied control. Let

$$\hat{u}_j(t) \triangleq u_j(t - \tau_{ij}(t)),$$

where $\tau_{ij}(t) \leq \bar{\tau}$ while (i, j) is maintained. If $\|\dot{u}_j(t)\| \leq \bar{\nu}$, then with $e_{ij}(t) \triangleq u_j(t) - \hat{u}_j(t)$, $\|e_{ij}(t)\| \leq \bar{e} \triangleq \bar{\nu} \bar{\tau}$.

Communication is local and unreliable: packets may be dropped, delayed, and received asynchronously. We model (i) probabilistic drops, (ii) bounded random delays, and (iii) independent broadcast schedules per robot, and report delivered bandwidth under these impairments in Sec. IV.

D. Problem Statement

Consider the multi-robot system above with dynamics (1) and communication model (7)–(8). Robots are advected by the spatially varying flow field $f_{\text{flow}}(x, t)$ and subject to bounded disturbances $d_i(t)$, which induce dispersion. While dispersion aids exploration, it can increase inter-robot distances and break communication links.

We target coverage-first exploration: during the exploration phase, dispersion is used to increase the sensed area while connectivity and safety are preserved.

Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ denote the physical disk graph induced by range $R_{\max}$ (Assumption 1). Robots may sparsify the network by maintaining a time-varying coordination subgraph $\mathcal{G}_{\text{topo}}(t) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(t))$ with $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$.

We encode safety and link-maintenance requirements using

$$\mathcal{C}_{\text{safe}} \triangleq \{x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \geq d_{\min}, \forall i \neq j\},$$

$$\mathcal{C}_{\text{conn}}(\mathcal{E}) \triangleq \{x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \leq R_{\max}, \forall (i, j) \in \mathcal{E}\},$$

and the admissible input set $\mathcal{U} \triangleq \{(u_1, \dots, u_N) : \|u_i\| \leq u_{\max} \forall i\}$ (Assumption 1).

Targets are not known a priori and are detected only when an agent enters sensing range. During exploration, the

team spreads to cover the environment (via flow-driven drift) while remaining connected and reducing communication redundancy; upon target detection, agents switch to goal-seeking while preserving end-to-end connectivity.

Each agent i accesses only local information: (i) its own state x_i ; (ii) relative states $\{x_j - x_i\}_{j \in \mathcal{N}_i(t)}$ and one-hop messages; and (iii) local graph quantities (e.g., two-hop connectivity) computable through neighbor exchanges. Let $\mathcal{I}_i(t)$ denote this information set. No centralized computation, global maps, or spectral quantities are assumed; the coordination structure is constructed distributedly from one-hop messages.

Design distributed policies consisting of (i) a feedback law $u_i(t) = \kappa_i(\mathcal{I}_i(t))$ for each agent and (ii) a concurrent pruning rule generating $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$, using only local information, such that for all $t \geq 0$:

- 1) $x(t) \in \mathcal{C}_{\text{safe}}$ and $x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_{\text{topo}}(t))$, and $\mathcal{G}_{\text{topo}}(t)$ remains connected.
- 2) The controller exploits advection and uses minimal actuation.
- 3) The team spreads for coverage and periodically constructs a sparse coordination topology using one-hop messaging; performance is evaluated via tree success rate, convergence time, and delivered bandwidth under impaired communication.

Sec. III provides a fully distributed realization via topology sparsification and decentralized control.

III. METHODOLOGY

Our framework couples a *graph layer* and a *control layer*. At discrete update times t_k , the graph layer constructs a sparse connected coordination structure using only one-hop neighbor messages and prunes all other links. Between updates, the control layer applies a CLF-CBF-QP, a nominal CLF drives task motion, while CBF constraints enforce collision avoidance and range maintenance on maintained coordination links. Although the disk graph $\mathcal{G}_t$ varies due to advection and diffusion, the range CBFs keep maintained links within $R_{\max}$ between updates, preventing their dropout.

Notation. Robots follow the control-affine model (3); the disk graph is defined in (7)–(8). Let t_k be the k th topology update time and $\mathcal{E}_{\text{topo}}(k) \subseteq \mathcal{E}_{t_k}$ the edge set (a spanning-tree coordination structure upon convergence). Define the maintained neighbor set $\mathcal{T}_i(k) \triangleq \{j : (i, j) \in \mathcal{E}_{\text{topo}}(k)\}$ and a separate collision-avoidance neighbor set $\mathcal{S}_i(t)$.

A. Distributed topology maintenance

We maintain a sparse coordination topology $\mathcal{G}_{\text{topo}}(t) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(t))$ as a connected subgraph of the instantaneous communication graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$, where $\mathcal{E}(t) = \{(i, j) : \|x_i(t) - x_j(t)\| \leq R_{\max}\}$ is induced by the communication radius $R_{\max}$. Topology updates are executed at discrete epochs; between epochs, the control layer enforces range constraints only on $\mathcal{E}_{\text{topo}}$.

Under underwater-style constraints, the topology layer must avoid exchanging full state per robot. We therefore

restrict messages to constant-size broadcasts and explicitly account for bytes/message and delivered bandwidth.

1) *Scalar hop to root propagation*: At the start of each topology update epoch, a designated root robot r initializes a scalar hop estimate $\delta_r = 0$, while all other robots initialize $\delta_i = \infty$. In all experiments, the root r is selected as the robot with the smallest ID at initialization and remains fixed for the duration of the run. Root failures and distributed re election are not studied in this paper. Each robot i periodically broadcasts the constant size tuple (i, δ_i, s_i) to its current one hop neighbors, where s_i is a local sequence counter. Upon receiving a message (j, δ_j, s_j) from neighbor j , robot i updates

$$\delta_i \leftarrow \min\{\delta_i, \delta_j + 1\}, \quad (9)$$

and records j as a candidate parent if the minimum is achieved. This is an online unweighted Bellman Ford Search (BFS) update over the time varying neighbor graph.

2) *Local parent selection and pruning*: After a fixed epoch duration T_{prune} , each robot $i \neq r$ selects a parent

$$p(i) \in \arg \min_{j \in \mathcal{N}_i(t)} \delta_j \quad \text{s.t.} \quad \delta_{p(i)} = \delta_i - 1, \quad (10)$$

Ties are broken by heuristics (e.g. smallest ID) The maintained edge set is then defined by the local keep rule:

$$\mathcal{E}_{\text{topo}} \triangleq \{(i, p(i)) : i \neq r, p(i) \text{ defined}\}. \quad (11)$$

Thus each robot maintains at most one outgoing topology edge; collectively, $\mathcal{G}_{\text{topo}}$ is a spanning tree like coordination structure *when the pruning epoch converges*. We quantify this event via the measured tree success rate and convergence time under impairments in Sec. IV.

Each broadcast contains constant-size fields (sender ID, δ_i , and sequence counter). We report (i) bytes per message, (ii) messages per topology update epoch, (iii) attempted (TX) and delivered (RX) bandwidth under packet drops and delays, and (iv) convergence time of the δ propagation procedure.

Robots broadcast asynchronously with jittered periods, and packets may be dropped or delayed. The implementation suppresses duplicates via sequence counters and terminates epochs based on observed stabilization. We quantify success rates (tree formation and root reachability) under these impairments in Sec. IV.

B. Illustrative example of a sparse graph

Fig. 1 illustrates δ -BFS pruning on a 6-node connected graph with root $r = 1$. The protocol propagates scalar hop-to-root estimates δ_i via constant-size broadcasts and then performs local parent selection to form a spanning-tree maintained topology. Each node maintains a *scalar* hop-to-root estimate δ_i (not all-pairs distances): initialize $\delta_1 = 0$ and $\delta_{i \neq 1} = \infty$, and periodically broadcast a constant-size message

$$m_i = (\text{id} = i, \delta_i, \text{seq}_i).$$

Upon receiving m_j from neighbor j , node i applies the local update

$$\delta_i \leftarrow \min\{\delta_i, \delta_j + 1\}. \quad (12)$$

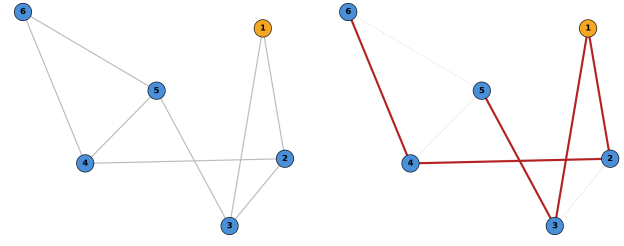


Fig. 1: Six-node example of δ -BFS pruning ($r = 1$). Left: initial communication graph $\mathcal{G}$. Right: maintained topology $\mathcal{G}_{\text{topo}}$ after local parent selection (red edges); dashed edges indicate pruned links.

For this example, the hop estimates converge to $\delta_1 = 0$, $\delta_2 = \delta_3 = 1$, $\delta_4 = \delta_5 = 2$, $\delta_6 = 3$.

Each node then selects a parent among neighbors one hop closer to the root,

$$C_i = \{j \in N_i : \delta_j = \delta_i - 1\}, \quad p(i) = \min C_i,$$

yielding $p(2) = 1$, $p(3) = 1$, $p(4) = 2$, $p(5) = 3$, $p(6) = 4$ and the maintained edge set

$$\mathcal{E}_{\text{topo}} = \{(1, 2), (1, 3), (2, 4), (3, 5), (4, 6)\},$$

Equivalently, keep $\{i, j\} \in E$ iff $p(i) = j$ or $p(j) = i$; the resulting spanning-tree coordination graph is used by the control layer to enforce range constraints only on maintained edges.

We evaluate robustness by injecting packet drops, bounded delays, asynchronous broadcasts, and additional channel impairments (Sec. IV), measuring success, convergence time, and delivered payload throughput.

C. Nominal control and safety constraints

After each topology update epoch yields the maintained edge set $\mathcal{E}_{\text{topo}}$ (Sec. III-A), we apply a standard CLF-CBF QP safety filter: a nominal goal-seeking command is minimally modified to satisfy collision avoidance and range constraints *only* on maintained links. Our contributions focus on the communication-feasible topology layer, not the CLF-CBF construction.

Given goal $g \in \mathbb{R}^3$, define

$$V_i(x_i) \triangleq \|x_i - g\|^2, \quad (13)$$

and generate a nominal controller u_i^{nom} (Sec. IVC). The CLF constraint is

$$\dot{V}_i(x_i) \leq -c_V V_i(x_i) + \epsilon_i, \quad (14)$$

with $c_V > 0$ and relaxation $\epsilon_i \geq 0$ for feasibility when safety constraints dominate.

To avoid attempting to preserve all links in the instantaneous dense neighbor graph, safety/communication constraints are enforced only on $(i, j) \in \mathcal{E}_{\text{topo}}$.

Range maintenance: for each $(i, j) \in \mathcal{E}_{\text{topo}}$, let

$$h_{ij}(x) \triangleq R_{\text{max}}^2 - \|x_i - x_j\|^2, \quad (15)$$

Algorithm 1: Comm-feasible δ -BFS pruning

Input : Neighbor set $\mathcal{N}_i$, root id r , T_{prune} , T_{stable}
Output: Local maintained parent $p(i)$ and edge $(i, p(i))$ if defined
State: $\delta_i \in \mathbb{N} \cup \{\infty\}$, $p(i) \in \mathcal{N}_i \cup \{\emptyset\}$, seq $s_i \in \mathbb{N}$, last-seq $\{s_j^{\text{last}}\}_{j \in \mathcal{N}_i}$, last-update time t_i^{upd}
Msg: $\langle i, \delta_i, s_i \rangle$
if $i = r$ **then**
 $\delta_i \leftarrow 0$
else
 $\delta_i \leftarrow \infty$
 $p(i) \leftarrow \emptyset$; $s_i \leftarrow 0$; $t_i^{\text{upd}} \leftarrow 0$; **foreach** $j \in \mathcal{N}_i$ **do**
 $s_j^{\text{last}} \leftarrow -1$
while $t < T_{\text{prune}}$ **and** $(t - t_i^{\text{upd}}) < T_{\text{stable}}$ **do**
 broadcast $\langle i, \delta_i, s_i \rangle$ to all $j \in \mathcal{N}_i$; $s_i \leftarrow s_i + 1$
 foreach received $\langle j, \delta_j, s_j \rangle$ **do**
 if $s_j > s_j^{\text{last}}$ **then**
 $s_j^{\text{last}} \leftarrow s_j$; // drop
 duplicates/out-of-order
 $\delta_{\text{new}} \leftarrow \min(\delta_i, \delta_j + 1)$
 if $\delta_{\text{new}} < \delta_i$ **then**
 $\delta_i \leftarrow \delta_{\text{new}}$; $t_i^{\text{upd}} \leftarrow t$
if $i \neq r$ **and** $\delta_i < \infty$ **then**
 $\mathcal{P}_i \leftarrow \{j \in \mathcal{N}_i : \delta_j = \delta_i - 1\}$
 if $\mathcal{P}_i \neq \emptyset$ **then**
 $p(i) \leftarrow \arg \min_{j \in \mathcal{P}_i} j$ // tie-break: smallest ID
return maintain $(i, p(i))$ if $p(i) \neq \emptyset$;
 $\mathcal{E}_{\text{topo}} = \{(i, p(i)) : i \neq r, p(i) \neq \emptyset\}$

and impose (using relative dynamics in Sec. II-B)

$$\dot{h}_{ij}(x) + \alpha_h h_{ij}(x) \geq 0, \quad \forall (i, j) \in \mathcal{E}_{\text{topo}}, \quad (16)$$

with $\alpha_h > 0$.

Collision avoidance: for nearby robots, define

$$h_{ij}^{\text{col}}(x) \triangleq \|x_i - x_j\|^2 - d_{\min}^2, \quad (17)$$

and enforce $\dot{h}_{ij}^{\text{col}}(x) + \alpha_c h_{ij}^{\text{col}}(x) \geq 0$ for $j \in \mathcal{N}_i^{\text{col}}$, where $\mathcal{N}_i^{\text{col}}$ is a short-range sensing set independent of $\mathcal{E}_{\text{topo}}$.

D. Systems implementation

At each control update, robot i solves a small QP that minimally modifies the nominal command to satisfy the CBF constraints on its incident maintained edges:

$$\begin{aligned} u_i^* \in \arg \min_{u_i, \epsilon_i} \quad & \|u_i - u_i^{\text{nom}}\|^2 + \rho \epsilon_i^2 \\ \text{s.t.} \quad & \dot{V}_i(x_i) \leq -c_V V_i(x_i) + \epsilon_i, \\ & \dot{h}_{ij}(x) + \alpha_h h_{ij}(x) \geq 0, \quad \forall j : (i, j) \in \mathcal{E}_{\text{topo}}, \\ & \dot{h}_{ij}^{\text{col}}(x) + \alpha_c h_{ij}^{\text{col}}(x) \geq 0, \quad \forall j \in \mathcal{N}_i^{\text{col}}, \\ & \|u_i\| \leq u_{\max}, \quad \epsilon_i \geq 0, \end{aligned} \quad (18)$$

where $\rho > 0$ weights CLF relaxation. This formulation is standard in the CLF CBF literature; we use it as a practical safety filter on the maintained topology.

The QP is derived using the design model in Sec. II B, while the experimental evaluation in Sec. IV evolves the robots under the advection diffusion model (Sec. II A) with packet drops/delays and neighbor churn. Thus the

results quantify closed loop robustness under the physics and communication impairments of interest, rather than relying on idealized synchronization assumptions.

IV. SIMULATIONS AND RESULTS

A. Simulation setup

We evaluate (i) synthetic stress tests that isolate the pruning message-passing protocol on random connected graphs under packet drops, bounded random delays, and asynchronous broadcasts, measuring TX/RX counts, delivered bandwidth, and convergence time; and (ii) end-to-end 3D HoloOcean simulations [21] with BlueROV2 vehicles [22] under advection–diffusion dynamics (Sec. II, (2)). In HoloOcean, robots are released together, drift and spread with the flow to expand sensing coverage, and then collectively reconfigure toward a detected goal/target, while asynchronous topology updates track neighbor churn in the disk graph (7)–(8) and the controller enforces range maintenance to preserve end-to-end connectivity.

a) *Baselines:* We compare four maintained-topology strategies. **Fully Connected** retains all instantaneous disk-graph edges (high connectivity, high communication/control cost). **Centralized MST (oracle)** recomputes a minimum spanning tree from a global disk-graph snapshot each epoch, giving an ideal edge-count lower bound but requiring non-local information (not one-hop implementable). **Adjacency Consensus** prunes locally by exchanging one-hop neighbor lists and removing an edge only when redundancy is certified by a two-hop witness, yielding conservative pruning and higher messaging. **Distributed Pruning (ours)** uses constant-size broadcasts (id, δ , seq) with local parent selection to form a spanning-tree-like maintained topology; we report bytes/message, messages/epoch, and delivered bandwidth under impairments.

Exact distributed MST construction is message-intensive and requires substantial coordination, which is poorly matched to underwater bandwidth limits and impairments. We therefore compare against an *adjacency-consensus* pruning baseline representative of practical local redundancy-removal: robots exchange one-hop neighborhoods and prune edges only with a two-hop witness. This captures the key tradeoff of conservative sparsification with variable-size payloads, and provides a meaningful comparison to our constant-size δ -BFS pruning.

TABLE I: Communication comparison

N	Method	Success (%)	Conv. time (s)	Payload (B/s)	Final edges
10	δ -BFS	100	5.68 ± 0.20	465 ± 25	9
10	Adj-Consensus	100	65.53 ± 36.52	25208 ± 2289	18
20	δ -BFS	100	8.42 ± 1.13	1972 ± 96	19
20	Adj-Consensus	100	208.02 ± 22.33	297805 ± 49986	27

Across our experiments, δ -BFS matches adjacency consensus in success rate (100%) while reducing convergence time by $\sim 11.5\times$ ($N=10$) and $\sim 36.6\times$ ($N=20$), lowering delivered payload rate by $\sim 54\times$ ($N=10$) and $\sim 151\times$

($N=20$), and yielding a sparser final graph (9 vs. 18 edges at $N=10$; 19 vs. 27 at $N=20$); see Table I.

B. Metrics

A topology update epoch is successful if the maintained graph $\mathcal{G}_{\text{topo}}$ is a connected spanning tree (equivalently, connected with $|\mathcal{E}_{\text{topo}}| = N - 1$).

- Tree success rate (%) over Monte Carlo trials under a given impairment setting.
- Time to first tree t_{tree} (s) is the first time the maintained topology becomes a connected spanning tree.
- Time to stable tree t_{stable} (s) is the first time the tree remains valid for T_{stable} seconds.
- TX/RX counts per epoch and delivered pruning payload throughput (B/s).
- Transmit queue delay p95 (enqueue-to-send) and overflow rate when applicable.

If an epoch does not converge, robots retain the last valid maintained topology for one epoch. If no prior valid topology exists, robots fall back to maintaining all current neighbor links for one epoch, then retry pruning.

We report the number of maintained edges $|\mathcal{E}_{\text{topo}}|$ and the edge reduction relative to the instantaneous disk graph:

$$\text{Edge reduction} = \frac{|\mathcal{E}(t_k)| - |\mathcal{E}_{\text{topo}}(k)|}{|\mathcal{E}(t_k)|} \times 100\%,$$

where $|\mathcal{E}_{\text{topo}}(k)| = N - 1$ by construction when the tree forms successfully.

For 3D HoloOcean runs we additionally report:

- **Goal-reaching time** or final distance to goal at horizon.
- **Connectivity preservation**: fraction of time the maintained topology remains connected during motion.
- **Safety**: minimum inter robot distance.

C. Communication accounting and stress tests

Each pruning broadcast contains a constant-size tuple (id, δ , seq) (6 bytes/message payload in our implementation, excluding transport overhead). Under lossy communication ($p_{\text{drop}} = 0.1$, $\tau_{\text{max}} = 0.5$, $j = 0.2$), the protocol converges in 9.27 s ($N=10$) and 9.85 s ($N=20$), delivering 95.5 and 239.4 B/s of received payload, respectively (Table II). Delivered throughput is RX payload bytes divided by the pruning-epoch duration (which may end early after T_{stable} with no δ updates); we report both TX and RX.

To strengthen realism, we incorporate three impairment models representative of underwater communication: (i) bursty packet loss via a two-state Gilbert-Elliott (GE) channel, (ii) bandwidth bottlenecks via a per-robot TX rate cap with a FIFO transmit queue, and (iii) distance-dependent impairments (Sec. IV-E). For each setting we measure: (i) tree formation success, (ii) time to first tree t_{tree} and time to stable tree t_{stable} (stability window T_{stable}), and (iii) TX/RX counts and delivered payload throughput.

Fig. 2 characterizes the correlated-loss structure induced by GE relative to iid drops using the parameterizations employed in our experiments. The CCDF of drop-burst lengths highlights how GE produces longer consecutive outages,

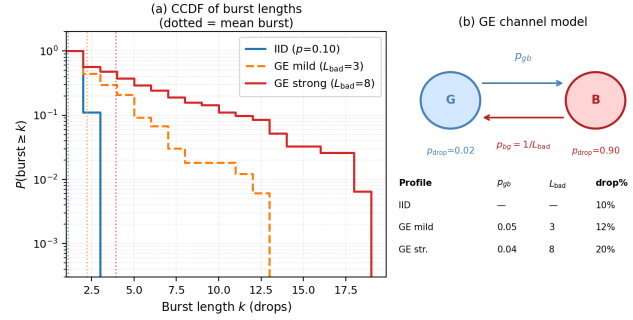


Fig. 2: Bursty-loss channel characterization. (a) CCDF of consecutive drop-burst lengths under iid loss and two GE parameterizations (mild/strong); dotted lines indicate mean burst length. (b) Two-state GE model and parameters used in our experiments.

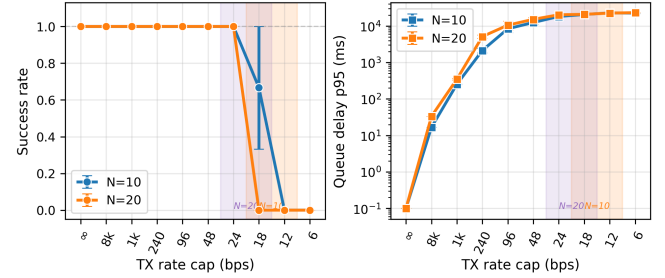


Fig. 3: FIFO rate limiting. Success and transmit-queue delay p95 versus TX rate cap for $N=10$ and $N=20$; breakpoints: 12–18 bps and 18–24 bps.

which can delay information propagation during pruning epochs.

Fig. 3 reports pruning feasibility under bandwidth bottlenecks for $N=10$ and $N=20$. As the TX rate cap decreases, transmit-queue delay p95 (enqueue-to-send) grows rapidly and pruning success collapses below a team-size-dependent breakpoint. Table II summarizes representative caps and the resulting feasibility metrics.

Distance-dependent impairments are evaluated by making drop probability increase with inter-robot distance and comparing weak versus strong dependence (Fig. 4).

D. Baseline comparison in lightweight simulation

We compare maintained-topology strategies in a lightweight simulator to illustrate sparsification dynamics and to provide a method-to-method baseline under controlled conditions. Realistic end-to-end validation in 3D HoloOcean under advection-diffusion currents and impaired communication is reported in Sec. IV-E.

Fig. 5 shows the maintained edge count $|\mathcal{E}_{\text{topo}}(t)|$ over time for a representative run. The fully connected strategy maintains all links, while the centralized MST (*oracle baseline*) immediately yields a tree with $N - 1$ edges. Our distributed pruning progressively reduces the maintained edge set toward a sparse, near-tree structure, whereas the sequential pruning baseline remains denser.

TABLE II: FIFO rate limiting summary, $T_{\text{prune}}=30$ s. “Runs” reports the number of seeds per row. “rx” is delivered pruning payload throughput (B/s).

N	Cap (bps)	Description	Runs	Success	t_{tree} (s)	Delay p95 (ms)	rx (B/s)
10	∞	No limit	10	1.00	2.9 ± 0.2	0	185
10	250	High cap	10	1.00	4.3 ± 0.1	2135 ± 359	169
10	48	2 $\times$ above breakpoint	10	1.00	11.5 ± 0.7	12483 ± 1336	38
10	30	Above breakpoint	10	1.00	18.0 ± 2.0	15094 ± 2288	18
10	18	Near breakpoint (partial)	10	0.67	27.8 ± 1.4	20700 ± 317	10
10	12	Below breakpoint (failure)	10	0.00	–	22939 ± 664	3
20	∞	No limit	10	1.00	3.0 ± 0.1	0	546
20	250	High cap	10	1.00	5.0 ± 0.1	4717 ± 274	405
20	48	2 $\times$ above breakpoint	10	1.00	15.0 ± 0.5	15261 ± 1891	67
20	30	Above breakpoint	10	1.00	20.6 ± 0.8	19439 ± 679	38
20	24	Near breakpoint (first success)	10	1.00	27.2 ± 0.9	20317 ± 584	27
20	18	Below breakpoint (failure)	10	0.00	–	20933 ± 591	13

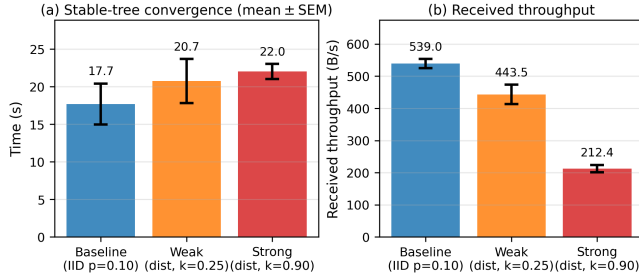


Fig. 4: Distance-dependent impairment stress test ($N=20$, 5 seeds). Stronger distance dependence increases stabilization time and reduces delivered throughput, while success remains high in this regime.

E. 3D HoloOcean validation under advection diffusion and impaired communication

We validate end to end operation in the 3D HoloOcean simulator with multiple BlueROV2 vehicles under flow driven dispersion. Robot motion evolves under the advection diffusion model (Sec. II) integrated via (2), which induces neighbor churn in the disk communication graph (7) (8). During motion, robots execute topology update epochs using the comm feasible δ BFS pruning protocol (Alg. 1) under packet drops, bounded random delays, and asynchronous broadcasts. The control layer then enforces range maintenance constraints while robots pursue a goal.

Fig. 6 shows a representative HoloOcean deployment under flow-driven dispersion. Fig. 7 reports the corresponding pruning epoch: the instantaneous disk-graph communication network is pruned to a spanning-tree maintained topology when the pruning epoch converges, and the overlay reports TX/RX counts and observed drops for that epoch.

V. CONCLUSION

We presented a fully distributed framework for connectivity-preserving deployment of flow-driven robot fleets under bandwidth-limited and unreliable communication. The topology layer constructs a spanning-tree coordination structure using constant-size neighbor broadcasts and local parent selection, and we explicitly quantify message budgets and delivered bandwidth under packet drops, random delays, and asynchronous broadcasts.

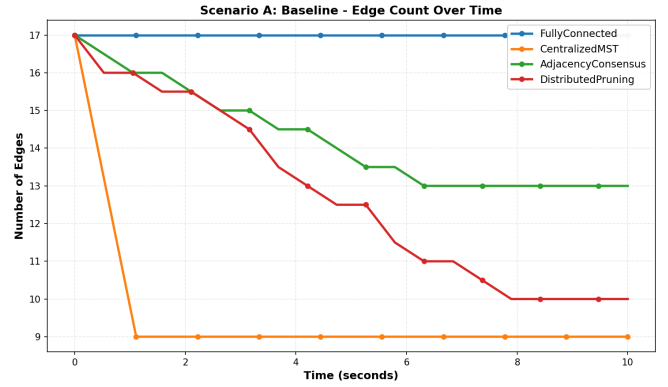


Fig. 5: Maintained edge count over time for four topology strategies.

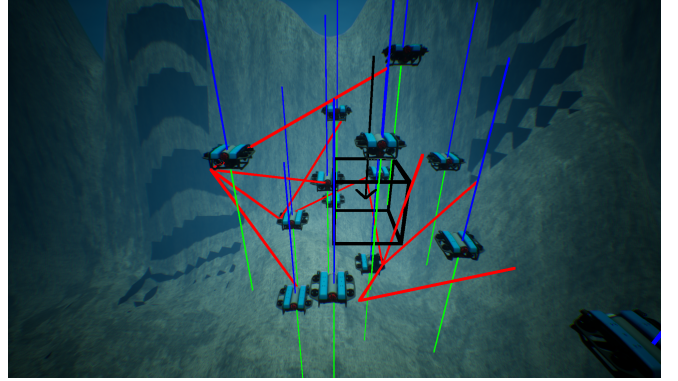


Fig. 6: 3D HoloOcean scene with multiple BlueROV2 vehicles during deployment under flow-driven dispersion.

On top of the maintained topology, each robot applies a standard CLF-CBF-QP safety filter that enforces collision avoidance and communication-range maintenance only on the maintained edges while executing goal-directed behavior under flow-driven dispersion.

We validated the approach through communication stress tests and 3D HoloOcean simulations with multiple BlueROV2 vehicles under an advection-diffusion motion model, demonstrating measured tree success rates and end-to-end connectivity with measured communication budgets.

Reported payload bytes exclude protocol overhead and therefore represent a lower bound on total channel usage. Communication impairments are modeled at the message level (drops, bounded random delays, and asynchronous broadcasts) rather than via a full acoustic modem model with packet-level effects and a realistic MAC stack.

Future work will (i) incorporate higher-fidelity acoustic modem and packet-level network models, (ii) scale to larger teams and longer-duration missions with repeated reconfiguration events, and (iii) validate the framework on field robotic platforms in real currents.

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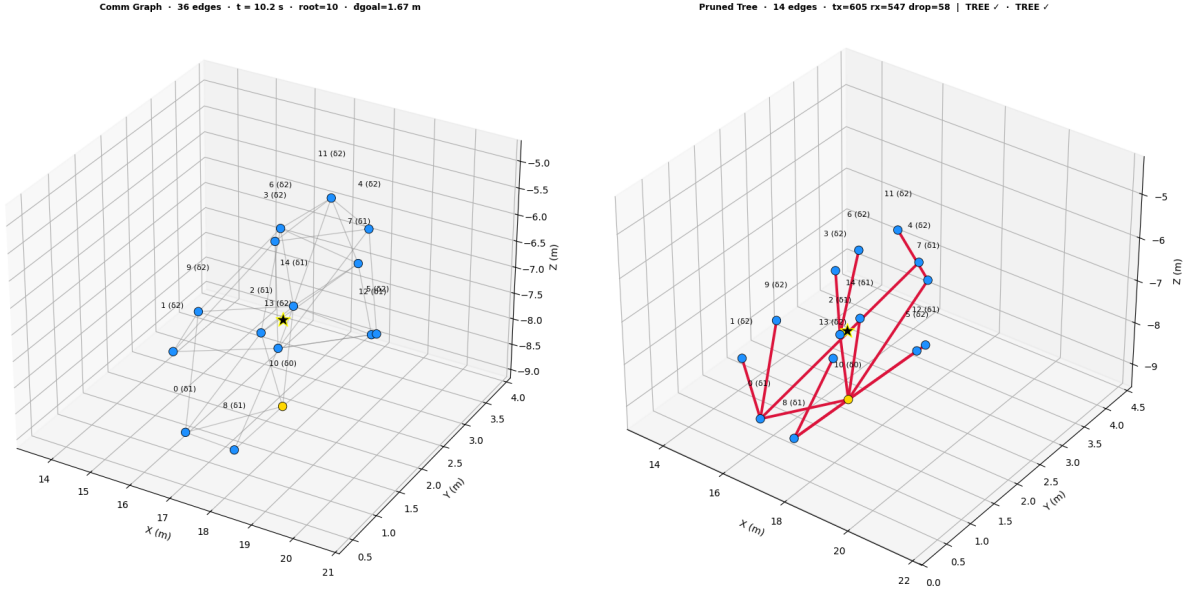


Fig. 7: HoloOcean pruning epoch ($N = 15$, $R_{\max} = 2.0$ m) under impaired communication ($p_{\text{drop}} = 0.1$). Left: instantaneous disk-graph communication edges. Right: pruned maintained topology $\mathcal{G}_{\text{topo}}$ produced by the online δ -BFS protocol (Alg. 1). The overlay reports TX/RX counts and observed drops for the epoch.

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